

STA2005: EXPERIMENTAL DESIGN: Tutorial 1

INTRODUCTION TO ED, RANDOMISATION

1. Suppose we wish to conduct an experiment with four treatments (A, B, C, D) each replicated three times. Twelve experimental units are available. The experimental units are fairly homogeneous.
 - Randomly assign the treatments to the experimental units using a random number table (Table A.1 in notes' appendix).
 - Describe in detail how you would do the randomisation drawing pieces of paper from a hat.
 - Any other suggestion (except using R, which we will leave for the practical)?
2. An experimenter wants to compare four treatments (A, B, C, D) and has 12 experimental units, but they are not homogeneous. However, they can be divided into 3 groups, each with four homogeneous units. What experimental design should be used? Use the table of random numbers to assign the treatments to the experimental units, and again carefully explain how you would do the randomisation using pieces of paper drawn from a hat.
3. Give two reasons why randomization is necessary in experimental design.
4. Read the following brief reports of statistical research, and identify:
 - (a) whether it was an observational study or an experiment.

If it was an observational study, identify (if possible)

- (b) whether it was retrospective or prospective. A prospective study is one in which subjects are identified in advance and one observes what happens to them over time, usually after an intervention. Retrospective studies identify subjects with a certain condition and then look at historical data to find associations.
- (c) the subjects studied, and how they were selected.
- (d) the parameter of interest.
- (e) the nature and scope of the conclusion the study can reach.

If it was an experiment, identify (if possible)

- (b) the subjects studied.

- (c) the factor(s) in the experiment, and the number of levels of each.
 - (d) the number of treatments.
 - (e) the response variable measured.
 - (f) the design.
 - (a) whether it was blind (or double-blind). A study is blinded if the patient never knows which treatment he/she is receiving. It is double-blind if also the doctor or the person applying the treatment and taking measurements does not know which treatment was assigned to which subject.
 - (g) the nature and scope of the conclusion the experiment can reach.
- (A) The leg muscles of men aged 60 to 75 were 50% to 80% stronger after they participated in a 16-week, high-intensity resistance-training program twice a week.
 - (B) Some people who race greyhounds give the dogs large doses of vitamin C in the belief that the dogs will run faster. Investigators tried three different diets in random order on each of five racing greyhounds. They were surprised to find that when the dogs ate high amounts of vitamin C they ran more slowly.
 - (C) In 2001 a report the *Journal of the American Cancer Institute* indicated that women who work nights have a 60% greater risk of developing breast cancer. Researchers based these findings on the work histories of 763 women with breast cancer and 741 women without the disease.
 - (D) Some gardeners prefer to use nonchemical methods to control insect pests in their gardens. Researchers have designed two kinds of traps, and want to know which design will be more effective. They randomly choose 10 locations in a large garden and place one of each kind of trap at each location. After a week they count the number of bugs in each trap.
 - (E) In a study appearing in the journal *Science*, a research team reports that plants in southern England are flowering earlier in the spring. Records of the first flowering dates for 385 species over a period of 47 years indicate that flowering has advanced an average of 15 days per decade, an indication of climate warming, according to the authors.
 - (F) Almost 90 000 women participated in a 16-year study of the role of the vitamin folate in preventing colon cancer. Some of the women had family histories of colon cancer in close relatives. In this at-risk group the incidence of colon cancer was cut in half among those who maintained a high folate intake. No such difference was observed in those with no family-based risk.

5. **Save the grapes.** Vineyard owners have problems with birds that like to eat the ripening grapes. Grapes damaged by birds cannot be used for winemaking (or much of anything else). Some vineyards use scarecrows to try to keep birds away. Others use netting that covers the plants. Owners really would like to know if either method works and, if so, which one is better. One owner has offered to let you use his vineyard this year for an experiment. Propose a design. Carefully indicate how you would set up the experiment, specifying the treatment and blocking factors and response variable.

Computer Practical 1

1. See `RforED.html` for examples on how to randomize treatments to experimental units using R.
2. Suppose you have 20 experimental units available and you have 4 treatments plus a control. You want to use a completely randomised design. Use R to randomise this experiment. In the end you should print out a table showing which experimental unit receives which treatment and in which order you will conduct the trials.
3. Now suppose you carry out another experiment exactly like the one above but this time you decide to block the experimental units into 4 blocks. Use R to randomise this experiment.
4. I have a 2^2 factorial experiment with type of music in 2 levels and loudness in 2 levels (response is ability to concentrate). I have 16 experimental units (all second year students). Use R to randomly assign treatments to experimental units in a) a completely randomised design, and b) a randomised block design where I am blocking for gender (8 males and 8 females).
5. Suppose I have an agricultural experiment with a field which I can divide into 5×5 plots. I have 5 treatments, each a different density of mielie plants (I am interested in how close together I should plant them for maximum yield). What design should I use? What are the experimental units? Use R to randomly assign treatments to experimental units.
6. **Categorical vs continuous variables:** Suppose, for an experiment I have a treatment factor temperature with levels 50, 100, 150 and 200 °C . Using the R code shown below, explain carefully how the parameter estimates and their meaning will differ when the following two linear models are fitted in R: a) temperature as a continuous variable, and b) temperature as a categorical variable.

- We first generate/simulate some data for the above fictional example:

```

ob1 <- rnorm(5, mean = 5, sd = 2)
ob2 <- rnorm(5, mean = 10, sd = 2)
ob3 <- rnorm(5, mean = 15, sd = 2)
ob4 <- rnorm(5, mean = 10, sd = 2)

# the above code generates data/observations/response values
# for the 4 treatment groups, 5 values for each group
# note all of the observations come from a normal distribution
# with the same standard deviation
# only the group means differ

# -- to keep these data (or any data) in a spreadsheet or a data file
#   we need a column for each variable

response <- c(ob1, ob2, ob3, ob4)           # response
treat <- rep(c(50, 100, 150, 200), each = 5) # treatment factor

```

- Make some plots of the data:

```

boxplot(response ~ treat, xlab = "temperature", ylab = "response")
plot(response ~ treat, xlab = "temperature", ylab = "response")

```

- Fit as a continuous variable:

```

m1 <- lm(response ~ treat)
summary(m1)
abline(m1)

```

- Temperature as a categorical variable:

```

temp <- as.factor(treat)

m2 <- lm(response ~ temp)
summary(m2)

```

Compare the two models with respect to parameter estimates, degrees of freedom, residual standard deviation, estimate for response when temperature = 100 °C. When would you use the one model, when the other?

In experimental design we almost always treat treatment factors as categorical.